IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-375

1. Topic (12 words max)

SMARTCINNAMON: A Multi-Modal Machine Learning and IoT Framework for Precision Cinnamon Cultivation

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-375
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Cinnamon (*Cinnamomum verum*) is one of Sri Lanka's most valuable agricultural exports, contributing significantly to rural livelihoods and foreign exchange earnings. However, the productivity, quality, and export competitiveness of the cinnamon industry are constrained by persistent challenges across the cultivation, post-harvest, and disease management stages.

At the cultivation level, soil nutrient management is still largely based on fixed-dose fertilizer schedules. This practice ignores spatial and temporal variations in soil fertility, often leading to over- or under-application of nutrients. The absence of real-time nutrient monitoring tools delays corrective action until visible deficiency symptoms appear, by which time yield potential is already compromised [1]. Precision nutrient management solutions using IoT sensors and machine learning could enable continuous monitoring, rapid detection of nutrient gaps, and site-specific fertilizer recommendations.

In the post-harvest stage, storage quality is a major risk factor. Cinnamon is highly sensitive to environmental fluctuations, and improper conditions -particularly high relative humidity (RH) and temperature -accelerate mold growth, degrade essential oil content, and reduce aroma quality. Beyond these, factors such as poor airflow, excess surface moisture, or the buildup of volatile organic compounds (VOCs) can silently compromise product integrity. Current monitoring systems are limited, tracking only basic parameters like temperature and RH, with no automated intelligence to detect multi-factor risks. A multi-parameter monitoring system that evaluates baseline ranges for temperature, RH, CO_2, O_2, VOCs, airflow, light exposure, dust, vibration, and condensation, combined with ML-driven risk analysis, could transform post-harvest management by enabling proactive intervention before quality degradation occurs [2].

Land-use inefficiency is another significant barrier to productivity. Many plantations contain underutilized or vacant areas due to irregular planting, plant loss, or poor replanting strategies. Manual surveys are labor-intensive and prone to human error, while the absence of spatial analytics prevents optimal replanting decisions and accurate yield forecasting [3].

Cinnamon plantations are also highly susceptible to foliar fungal diseases such as Cylindrocladium leaf spot and Anthracnose. These pathogens spread rapidly under humid conditions, causing defoliation, yield loss, and long-term plant stress. Current detection methods rely on visual inspection, which is subjective, slow, and lacks severity quantification. Without timely intervention, infections can spread widely before effective control measures are implemented [4]. Addressing these interconnected challenges requires an integrated, technology-enabled approach. By combining IoT-based soil monitoring, AI-powered post-harvest quality control, GIS-driven land optimization, and mobile-based fungal disease detection, the industry can significantly improve yield consistency, reduce losses, meet export compliance standards, and enhance economic sustainability.

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6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed *SMARTCINNAMON* platform is a unified, web-based intelligent and fully integrated digital decision-support system designed to support medium- and large-scale cinnamon growers and exporters by addressing critical gaps in cultivation, post-harvest quality control, land utilization, and disease management through the integration of IoT, GIS, AI, machine learning, and image processing technologies.

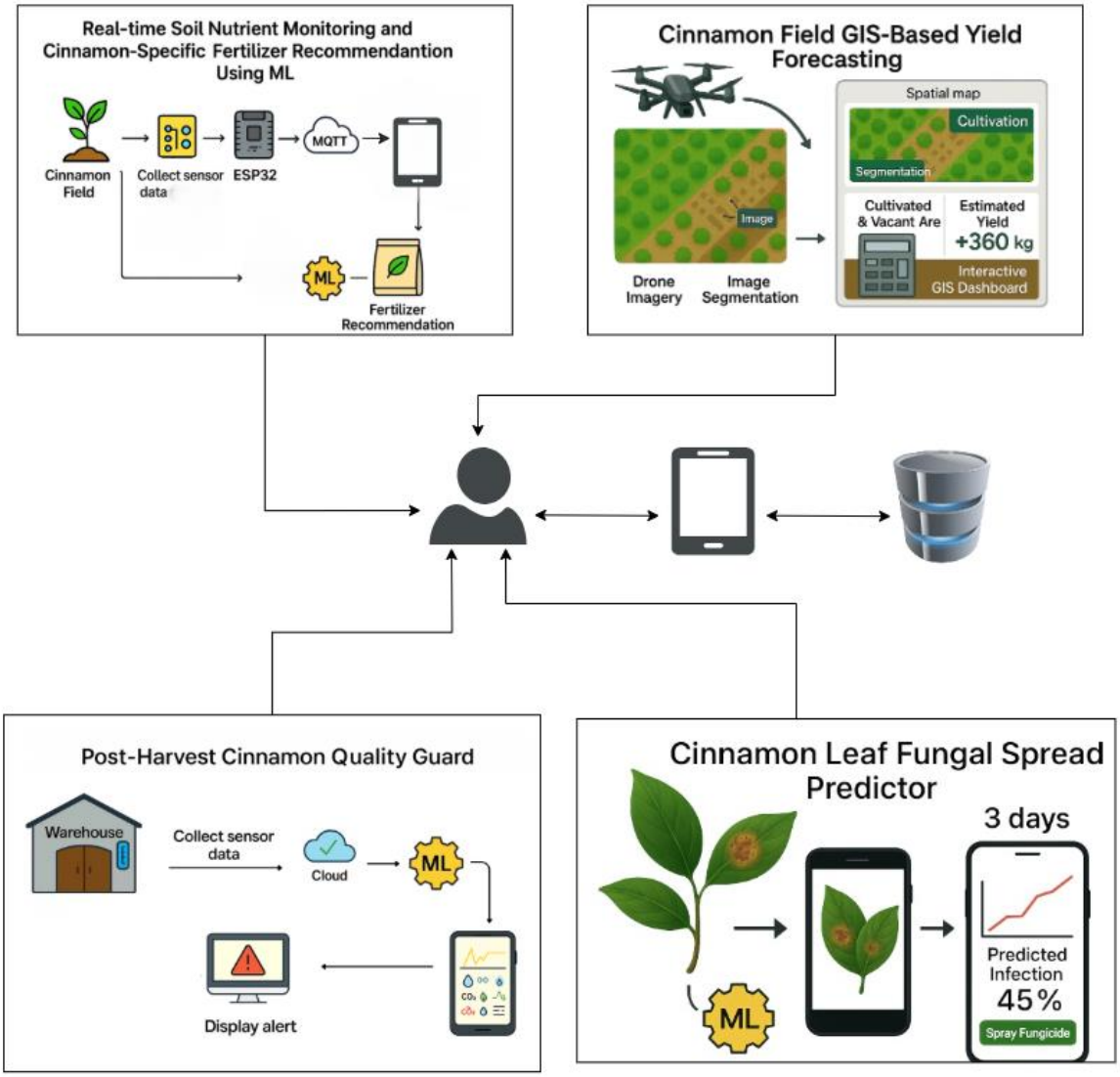
The **Cinnamon-Specific Fertilizer Deficit System** uses IoT-based NPK, moisture, and temperature sensors to continuously monitor soil conditions. Data is transmitted via ESP32 and MQTT to a cloud dashboard, where ML models calculate nutrient deficits and recommend precise fertilizer quantities, enabling precision fertilization instead of fixed-dose application.

The **Post-Harvest Cinnamon Quality Guard** employs a multi-sensor array to track 10 baseline-driven environmental parameters (temperature, relative humidity, CO_2, O_2, VOCs, light intensity, airflow, dust, vibration, and surface moisture). AI models compare live data against optimal baselines to detect mold/aroma risks and generate real-time corrective alerts, helping maintain Ceylon cinnamon's export-grade quality.

The **Cinnamon GIS Yield Planner** processes drone or satellite imagery with image segmentation and ML classification to map cultivated vs. vacant areas. It calculates available planting space, estimates plant counts, and forecasts yields, producing interactive maps for optimized replanting decisions.

The **Cinnamon Leaf Fungal Spread Predictor** is a mobile-based system that detects fungal infections on cinnamon leaves and predicts their future spread using CNN/MobileNet for detection and LSTM/GRU for time-series forecasting. Farmers capture a leaf photo, and the app estimates the current infection percentage and forecasts the future spread (e.g., in 3 days), providing early action advice such as spraying or pruning.

These modules connect to a unified dashboard accessible via mobile or web, delivering real-time insights, predictive analytics, and compliance documentation. The conceptual framework integrates field sensors, aerial imaging, AI models, and mobile tools into a seamless ecosystem that improves productivity, quality, and sustainability across the cinnamon value chain.


Real-time Soil Nutrient Monitoring and Cinnamon-Specific Fertilizer Recommendantion Using ML
Cinnamon Field
Collect sensor data
ESP32
MQTT
ML
Fertilizer Recommendation
Cinnamon Field GIS-Based Yield Forecasting
Spatial map
Cultivation
Segmentation
Image
Cultivated & Vacant Are
Estimated Yield
+360 kg
Interactive GIS Dashboard
Drone Imagery
Image Segmentation
Post-Harvest Cinnamon Quality Guard
Warehouse
Collect sensor data
Cloud
ML
Display alert
Cinnamon Leaf Fungal Spread Predictor
3 days
ML
Predicted Infection
45%
Spray Fungicide

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of the *SMARTCINNAMON* platform requires interdisciplinary expertise across agriculture, plant pathology, IoT engineering, machine learning, remote sensing, and post-harvest technology to ensure scientific accuracy, scalability, and practical field adoption.

From an **agronomy and soil science** perspective, knowledge of cinnamon growth cycles, nutrient uptake patterns, and irrigation management is vital. Specialists must define baseline NPK thresholds, soil pH, and moisture levels, and interpret live sensor data for site-specific fertilizer recommendations.

Post-harvest technology and quality assurance expertise is required to design multi-parameter environmental monitoring for cinnamon storage. This includes setting baselines for ten key factors (temperature, relative humidity, CO_2, O_2, VOCs, light intensity, airflow, dust, vibration, and condensation) and applying AI to detect deviations that trigger aroma loss or mold growth.

Remote sensing and GIS experts contribute by acquiring drone/satellite imagery, performing image segmentation, and classifying cultivated vs. vacant areas. Their expertise supports plant count estimation, replanting layout optimization, and accurate yield forecasting.

Plant pathology specialists are essential to identify diseases such as *Cylindrocladium* and *Anthracnose*, validate computer vision outputs, and standardize severity metrics. Their role extends to supporting time-series datasets for predicting the future spread of infections.

IoT engineers integrate multi-sensor hardware (NPK, RH, VOCs, etc.) with ESP32 controllers, establish MQTT/cloud pipelines, and ensure robust field connectivity.

Machine learning and computer vision specialists develop and optimize CNN, MobileNet, YOLOv9, LSTM/GRU, Random Forest, and CatBoost models for nutrient monitoring, GIS mapping, fungal detection, and spread forecasting. Their work includes dataset augmentation, model compression, and TFLite optimization for mobile deployment.

Key data requirements include:

- Annotated cinnamon leaf datasets (healthy, nutrient-deficient, fungal-infected, with time-series spread labels).
- Soil sensor datasets (NPK, pH, EC, moisture, temperature) linked to yield outcomes.
- Drone/satellite imagery labeled for cultivated vs. vacant zones.
- Baseline storage environment datasets (10 parameters).
- Historical climate and yield data for predictive modeling.

Integrating these specialized domains with high-quality datasets will ensure SMARTCINNAMON delivers reliable, proactive decision support for sustainable cinnamon production.

8. Objectives and Novelty

Main Objective

SMARTCINNAMON is an integrated AI, IoT, GIS, and machine learning–based system that enhances cinnamon production, post-harvest quality, and disease management. It includes real-time soil nutrient monitoring with precision fertilizer recommendations, post-harvest environmental monitoring for mold/aroma control with audit-ready logs, GIS-based field utilization and yield forecasting for optimized replanting, and mobile-based fungal disease detection with severity and spread estimation — improving yield, quality, compliance, and sustainability across the cinnamon value chain.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Bangamuwage.M.D-IT22592606	Real-time soil nutrient monitoring and cinnamon-specific fertilizer recommendation using ML (Cinnamon-Specific Fertilizer Deficit System)	- Deploy IoT sensors (JXBS-3001 for NPK, FC-28 for moisture, DHT11 for temperature & humidity) in cinnamon fields - Program ESP32 microcontroller and transmit data via MQTT - Display live readings in a cloud-based dashboard - Train ML models (Random Forest, CatBoost) on soil + cinnamon yield data - Use live NPK input to recommend fertilizer type and quantity - Compare real-time sensor data with ideal NPK thresholds for cinnamon - Calculate deficit in each nutrient (%) and recommend how much more fertilizer to apply.	**Intelligent Fertilizer Deficit Calculator for Cinnamon.** precision-fertilization tool tailored to cinnamon- most existing fertilizer recommendations in Sri Lanka use fixed-dose schedules, which often result in overuse or underuse of nutrients.Real-time nutrient gap analysis using IoT sensors, providing an instant percentage-based deficit for each nutrient (N, P, K) instead of waiting for periodic lab reports. Dynamic fertilizer dose calculation that converts nutrient deficit into exact fertilizer amounts (e.g., 25 kg Urea/acre) for immediate field application.Integration of ML models (Random Forest, CatBoost) trained on cinnamon-specific soil–yield datasets, making recommendations more

			accurate over time.Cloud-based dashboard for continuous monitoring and decision support, enabling data-driven nutrient management across multiple plantation sites.
Vihani W.W.G.D-IT22594372	To develop a post-harvest cinnamon "quality guard" system that continuously monitors multiple environmental parameters in warehouses, detects mold/aroma risks, and provides real-time corrective alerts.	– Install multi-sensor arrays at strategic warehouse points (door, center, corners). – Stream readings every 5–10 minutes to a central dashboard. – The system specifically monitors 10 parameters: 1) Temperature, 2) Relative Humidity (RH), 3) CO_2 concentration, 4) O_2 levels, 5) VOCs (Volatile Organic Compounds), 6) Light intensity, 7) Airflow speed, 8) Dust/particle levels, 9) Vibration/shock, 10) Surface moisture (condensation). – Define baseline thresholds for each parameter (e.g., RH 55–65%, Temperature 20–25°C, VOC < defined limit). – Apply ML models to compare live sensor data against baselines and detect deviations that may cause mold growth degradation. – Send real-time alerts with corrective action guidance (e.g., increase ventilation, activate dehumidifier, reduce light exposure). – Display storage trends via cloud/mobile dashboard for continuous monitoring.	**Cinnamon-specific post-harvest monitoring system** Goes beyond basic temp/RH tracking by analyzing 10 baseline-driven environmental factors. Unlike simple monitoring systems, this approach uses baseline comparisons + ML analytics to identify risk conditions early. This novelty ensures reduces mold development, and maintains Ceylon cinnamon's export-grade quality standards in a proactive, data-driven way.

Hansajith W.R.N IT22565594	Optimize cinnamon field utilization using drone imagery and GIS-based yield forecasting (cinnamon field GIS-based system)	- Acquire drone or satellite imagery of cinnamon plantation blocks - Apply image segmentation and machine learning to classify "cultivated" vs. "vacant" areas - Calculate available space and simulate replanting layout based on spacing (e.g., 2m × 2m) – Estimate plant count and project yield using tree count + historical bark productivity data. - Display a spatial map and interactive dashboard: "X new trees can be added → Estimated Yield +Y kg"	**Cinnamon-Specific GIS Yield Planner GIS-based vacant area detection for cinnamon plantations** eliminates manual surveying, which is labor-intensive and error-prone. Optimized replanting layouts are generated automatically using standard plantation spacing (e.g., 2m × 2m) to maximize land-use efficiency. Integrated yield forecasting uses both tree count and historical bark productivity to accurately predict revenue potential. The system calculates exact plant numbers for replanted areas, allowing precise long-term production planning. Being a drone/satellite-based solution, no physical soil or tree sensors are required, reducing cost. Interactive spatial maps and dashboards visually guide managers on where and how much to replant for maximum output.
Randeni R.P.R.M- IT22578778	Predict the future spread of fungal infection on cinnamon leaves using mobile image input and machine learning. (Mobile-Based Cinnamon Leaf Fungal Spread Predictor)	– Capture images of infected cinnamon leaves at different infection stages over time. – Preprocess images (resize, denoise, segment infected regions). – Annotate images with infection spread percentage and timeline. – Extract time-series features from image sequences. – Train a time-series forecasting ML model	**Cinnamon-Specific Fungal Spread Predictor** Mobile-based system that not only detects fungal infection on cinnamon leaves but also predicts its future spread rate using time-series ML models (LSTM/GRU). Unlike conventional detection tools that only identify current disease, this solution provides advance warnings of how

		(e.g., LSTM, GRU) to predict infection growth rate. – Input: Single mobile image + past data (optional). – Output: Predicted infection percentage after given time (e.g., in 3 days). – Visualize infection prediction and suggest early actions via mobile app.	infection will progress over the next few days. Farmers can take early preventive actions (spraying, pruning, isolating) before severe damage occurs, reducing yield loss and improving long-term crop health.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Bangamuwage.M.D-IT22592606	**Real-Time Soil Nutrient Monitoring & Cinnamon-Specific Fertilizer Deficit System** This component applies **IoT engineering and machine learning specialization** by integrating multi-parameter soil sensors (NPK, moisture, temperature, humidity) with an ESP32 microcontroller programmed for MQTT data transmission. A cloud-based dashboard displays real-time readings, while Random Forest and CatBoost ML models trained on cinnamon-specific soil–yield data generate precise fertilizer recommendations. The specialization is demonstrated through the full IoT hardware–software integration, sensor calibration, cloud connectivity, and predictive analytics model development tailored to cinnamon agronomy.
Vihani W.W.G.D-IT22594372	**Post-Harvest Cinnamon Quality Guard** This component applies **IoT and AI specialization** by integrating multi-sensors to monitor 10 baseline-driven parameters (temperature, RH, CO_2, O_2, VOCs, light, airflow, dust, vibration, surface moisture). ML models compare live readings against baselines to detect risks and guide corrective actions, ensuring proactive storage quality management for Ceylon cinnamon.
Hansajith W.R.N IT22565594	**Cinnamon Field GIS-Based Yield Planner** This component uses **remote sensing, GIS, and machine learning specialization** to process drone/satellite imagery for plantation management. Image segmentation and ML classification differentiate cultivated from vacant land. GIS algorithms calculate available planting space, estimate plant counts, and forecast yields using historical productivity data. The specialization is evident in spatial data processing, geospatial modeling, and yield prediction techniques that eliminate manual surveying and enable optimized land-use planning.
RANDENI R.P.R.M-IT22578778	**Mobile-Based Cinnamon Leaf Fungal Detection & Spread Prediction** This component applies mobile-based computer vision and time-series machine learning specialization to forecast the progression of fungal infections on cinnamon leaves. Using smartphone-captured images, the system preprocesses leaf data (resize, denoise, segment) and extracts infection-related features. Annotated infection stages form a temporal dataset for LSTM/GRU models, enabling accurate prediction of infection growth over a defined future period (e.g., 3 days). The mobile app outputs both the current infection percentage and a forecasted spread rate, along with early treatment suggestions.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.